

Analysis of COVID-Related Death Using Random Forest Classification

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I. Introduction

The rise of the COVID-19 pandemic took the healthcare field by surprise and at its peak was responsible for or a contributing factor to the death of over 460,000 U.S. citizens in 2021 (U.S. Centers for Disease Control and Prevention, 2026). While the presence of vaccines and an understanding of the virus has significantly reduced its impact on mortality, the virus has still been a contributor to or the sole cause of death in over 4,000 U.S. citizens in the first three months of 2026 (U.S. Centers for Disease Control and Prevention, 2026). With the virus still being responsible for a large number of deaths, it is important to understand the mortality risks associated with contracting the virus and what factors contribute to an increased probability of mortality.

As a result of increased COVID surveillance, large datasets have been collected, enabling machine learning algorithms to address these concerns. Methods such as random forest classification can be used to learn the relationships between demographic and clinical features and COVID-related mortality. By further analyzing the features most responsible for the construction of a random forest model, risk factors can also be hypothesized.

This analysis's primary goal was to create a random forest model tuned to most accurately identify deaths given demographic and healthcare-related features and to determine potential risk factors associated with COVID-related death. The model was tuned in such a way as to avoid misclassifying individuals who died as survivors. In a situation such as this, correctly

predicting who will die is the most important goal so that pre-emptive care can be given to minimize the probability of this undesired outcome.

Data were collected by the Centers for Disease Control and Prevention (CDC) as part of a COVID-19 case surveillance process. Observations represent a deidentified person who contracted the COVID-19 virus, and public health jurisdictions submitted these through a standardized case reporting form through July 5, 2024, the end of the reporting period (U.S. CDC, 2025). In total, there were roughly 106 million observations and 19 features, including death status, contained within the data. These features included a patient's demographics, location, and COVID-related health information——this includes the presence of underlying health conditions, whether a patient was admitted to an intensive care unit (ICU), whether a person was hospitalized, a patient's symptom status, and more.

Within most of these features, there were numerous instances of the missing and unknown categories, as well as empty values; however, each of these values had a different meaning. The missing value indicated case report form questions that did not include a response, the unknown value was an option on the case report questions, and empty values were used to suppress true values for patient privacy protection.

II. Exploratory Analysis and Preprocessing

Because of the uniquely large size of the COVID surveillance data, preprocessing steps were implemented alongside exploratory analysis to reduce the working size of the data and the computational expense associated with searching over the entirety of the data. The only exploratory analysis conducted over the entirety of the data was the calculation of category frequencies for each of the features. Because some features had upwards of 90 percent of their

values suppressed or missing, these features were immediately dropped. Additionally, we wanted our final model to be generalizable, meaning features related to the location or date of covid contraction were not of interest, so they too were dropped.

Upon removal of these features, the modeling data consisted of 8 categorical, explanatory features and the target feature. These features and their category levels are explained in Table 1.

Table 1: Features and Feature Categories

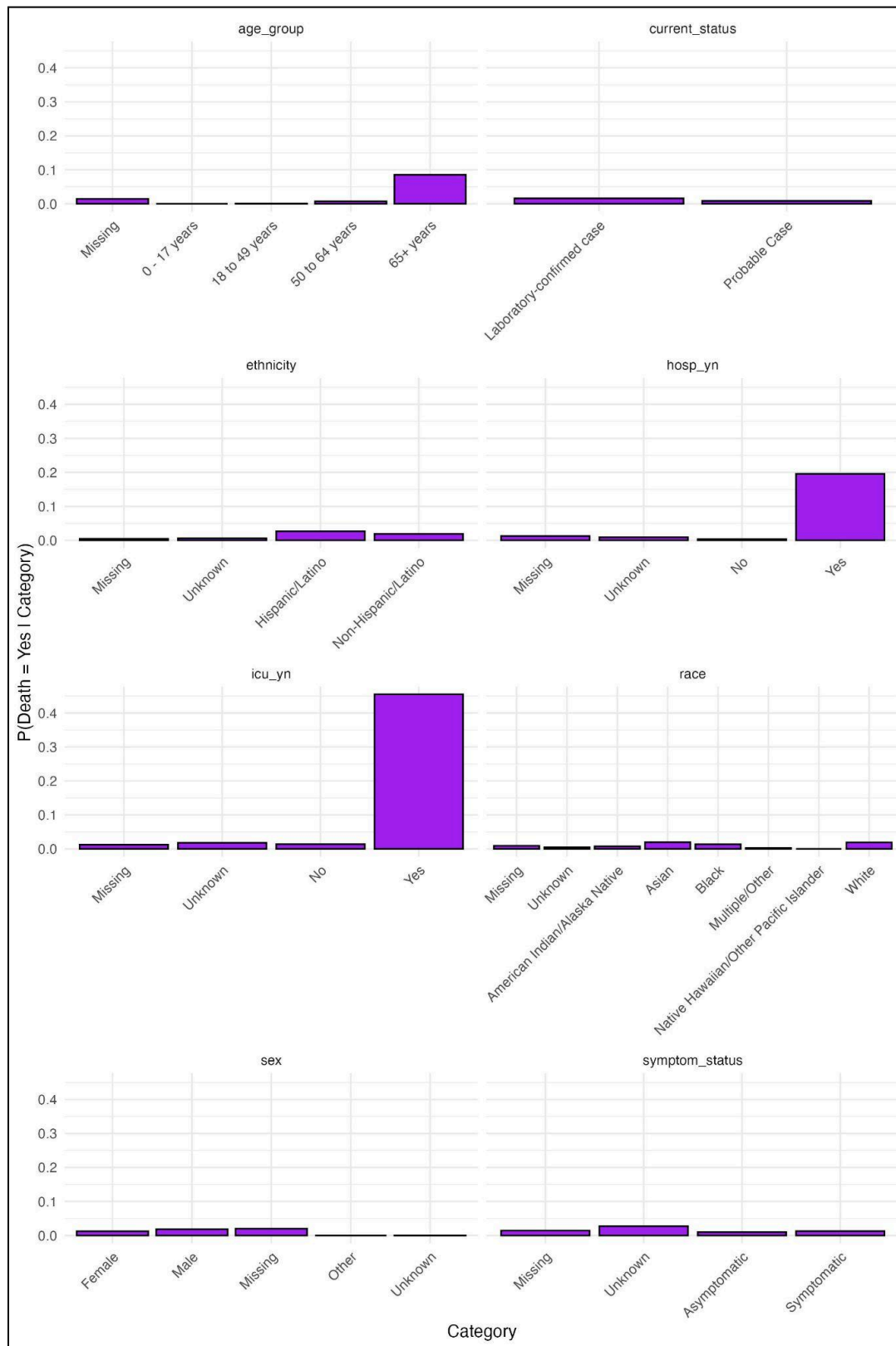
Feature	Description	Values
Death_yn (target)	Did the patient die as a result of this illness?	Yes, No, Unknown, Missing, NA
sex	Patient's sex	Female, Male, Other, Unknown, Missing, NA
race	Patient's race	American Indian/Alaska Native, Asian, Black, Multiple/Other, Native Hawaiian/Other Pacific Islander, White, Unknown, Missing, NA
ethnicity	Patient's ethnicity	Hispanic, Non-Hispanic, Unknown, Missing, NA
age_group	Age group in which the patient belongs	0 - 17 years, 18 - 49 years, 50 - 64 years, 65 + years, Unknown, Missing, NA
current_status	What is the current status of this patient's illness?	Laboratory-confirmed case, Probable case
symptom_status	What is the symptom status of this patient?	Asymptomatic, Symptomatic, Unknown, Missing
hosp_yn	Was the patient hospitalized?	Yes, No, Unknown, Missing
icu_yn	Was the patient admitted to an intensive care unit (ICU)?	Yes, No, Unknown, Missing

When analyzing our target feature, a categorical death status indicator, we found that this too was filled with missing, unknown, and empty values. Because we were concerned with estimating the death or survival of a patient, we only kept the subset of observations for which the death status was true or false, corresponding to the target feature categories “Yes” and “No.” This left just under 32 million observations for model creation. Following this, we found that there was severe class imbalance between the death status categories—the majority class, patients who survived, contained approximately 67.5 times the observations as the minority class, deceased patients.

To better understand which features or feature categories may prove useful in prediction, the conditional probability of death for every category within every feature was calculated, and the results are displayed by feature in Figure 1. By calculating the conditional probability of death, or the proportion of patients who died given each feature category, we were able to determine that certain feature categories resulted in a much higher probability of death than others. Based on the conditional probabilities, we observed that patients in the 65 or older age group had a significantly higher probability of death than those in other groups. Additionally, patients who had been hospitalized or admitted into an intensive care unit had a significantly higher probability of death than those who had not been hospitalized or placed in intensive care. These findings built a foundation for possible risk factor determinations.

From our exploratory analysis, we found that the conditional probability of death given a category of missing or unknown was slightly elevated compared to other categories in some features. Because of this, we noted that missingness and unknown values could be predictive of death, and these were left as categories for random forest creation. Because empty values indicated suppression, this was no different from not having the information from the case report,

Figure 1: Probability of Death Conditional on Feature Categories



causing us to merge the empty values into the missing category. Ultimately, we preprocessed features to maintain missingness and unknown case report answers as potentially predictive categories of death.

Because there were very few observations with empty values in the age group feature, we instead decided to mode impute the empty values. While mode imputing before performing cross-validation for model selection may appear to cause data leakage, invalidating the optimal model from the cross-validation process, the 17-49 year old age group dominated the feature, thus ensuring that mode imputation would result in empty values being populated with this age group regardless of the data splits used in cross-validation. This means mode imputation would not provide much, if any, data leakage during the cross-validation process.

Initially, we attempted to create our random forest classifier using all of the nearly 32 million observations, but the computational expense was far greater than our machines could handle. Due to this and our imbalanced classes, we decided to undersample 50,000 observations from both the majority and minority classes and combine them to create a smaller, balanced training set with 100,000 observations. Before this undersampling, we needed to first split the nearly 32 million observation data set into an initial train and test set and perform the undersampling from the initial training set to guarantee that our test set would not contain observations shared by the undersampled training data. This initial split was performed using class stratification so that the test set maintained the true class imbalance observed from the data, enabling us to test our model on data more closely representative of the true distribution of the death status feature. The test set had one percent of the nearly 32 million observations, giving a test set of just above 319,000 observations.

III. Models

Although the primary goal of our analysis was to develop a model optimized to correctly identify individuals who died from COVID, a second random forest was created to optimize overall predictive accuracy for comparison. Additionally, a naive baseline model was implemented to provide a reference point for evaluating model performance. The naive model's purpose was to demonstrate the predictive ability of the simplest possible model and provide a benchmark for the random forest models. Because the testing data were highly class-imbalanced, the naive model would produce the best results by simply predicting that each patient in the test set survived.

Both random forest models were selected using five-fold cross-validation on the balanced, undersampled training data over a hyperparameter grid that included multiple values for the maximum number of features considered at each split in the underlying decision trees. While the minimum number of observations required in the terminal nodes of underlying trees is a valid hyperparameter to consider, we fixed this value at one and did not tune it. If the underlying trees were to become too complex as a result of this fixed hyperparameter, then our maximum number of features hyperparameter would result in a lower optimal value that would reduce the overfitting from the fixed terminal node hyperparameter. To ensure that the primary model prioritized correctly identifying those who died, we selected the cross-validated model with the highest sensitivity. The individual trees comprising the random forest models were constructed using Gini impurity as the splitting criterion.

Our second goal was to determine feature categories that may be associated with an increased risk of COVID-related death using available data. This was achieved using the

permutation feature importance (PFI) scores from the random forest model optimized for sensitivity in conjunction with the conditional probabilities of death shown in Figure 1.

PFI scores measure how important a feature is to predictive performance. Because PFI scores are calculated after model training, categorical features are scored at the encoded category level. First, a model is trained and tested to obtain a classification accuracy. Then, for each encoded feature, the given feature's values are shuffled to break the relationship with the response. The classification accuracy is again collected for this shuffled data, and the change in accuracy from the baseline model is calculated—this value is the PFI score. Larger changes in accuracy indicate a greater predictive importance of the feature (Chan, 2026).

Because PFI reflects predictive importance rather than a causal relationship with the outcome, it cannot be used to determine risk factors. To provide supporting evidence for whether a feature category may be a risk factor, we compared PFI scores with the conditional probability of death for each category. Feature categories that demonstrated both high predictive importance and elevated conditional probabilities of death were interpreted as potential risk factors. While this introduced a way to determine potential risk factors, the presence of high multicollinearity between features can result in PFI scores being partially attributed to the wrong feature (Chan 2026).

IV. Model Results

By applying the naive model decision rule to each of the test observations, this model achieved an accuracy of 98.53 percent. In many circumstances, this model's accuracy would outperform even some of the most complex models. Since the naive model achieves its highest possible accuracy by predicting that every patient survives, this perfectly highlights the issue of

modeling for accuracy in a healthcare setting. The model trades off the ability to correctly identify any deaths to obtain its high accuracy, resulting in all deceased patients being misclassified as survivors. This result motivates the use of a more advanced machine learning technique, such as the random forest classifier.

After both random forest models were created, they were evaluated on the test data of just over 319,000 observations, which maintained the original class imbalance. While both models resulted in high values for their respective optimization metrics, the differences in the metrics from both models were not substantial.

The accuracy-based random forest model achieved an accuracy of 87.39 percent, which is relatively high. While the naive model achieved an accuracy much higher than this by predicting each patient as a survivor, the naive model failed to correctly identify any patients who actually died. Conversely, the random forest model correctly predicted 95.92 percent of patients who died, making the random forest model far superior to the naive model in a healthcare context. The confusion matrix of prediction results is viewable in Figure 2.

Figure 2: Accuracy-Based Random Forest Confusion Matrix

Predicted	Yes	40,053	4,487
	No	274,533	191
		No	Yes
		Actual	

Because the primary random forest model was optimized to correctly identify those who died, it is expected that the model would trade some accuracy for an increase in sensitivity.

While this ultimately happened, the tradeoff was minimal. The primary random forest model achieved an accuracy of 86.99 percent, only 0.4 percentage points worse than the accuracy-optimized model. This model also achieved a sensitivity of 96.30 percent, 0.38 percentage points larger than the accuracy-optimized model. The primary model achieved these metrics by predicting that just over 1,200 more patients from the test observations died than the accuracy-optimized model. This can be seen by comparing the confusion matrices of prediction results in Figure 2 and Figure 3.

Figure 3: Sensitivity-Based Random Forest Confusion Matrix

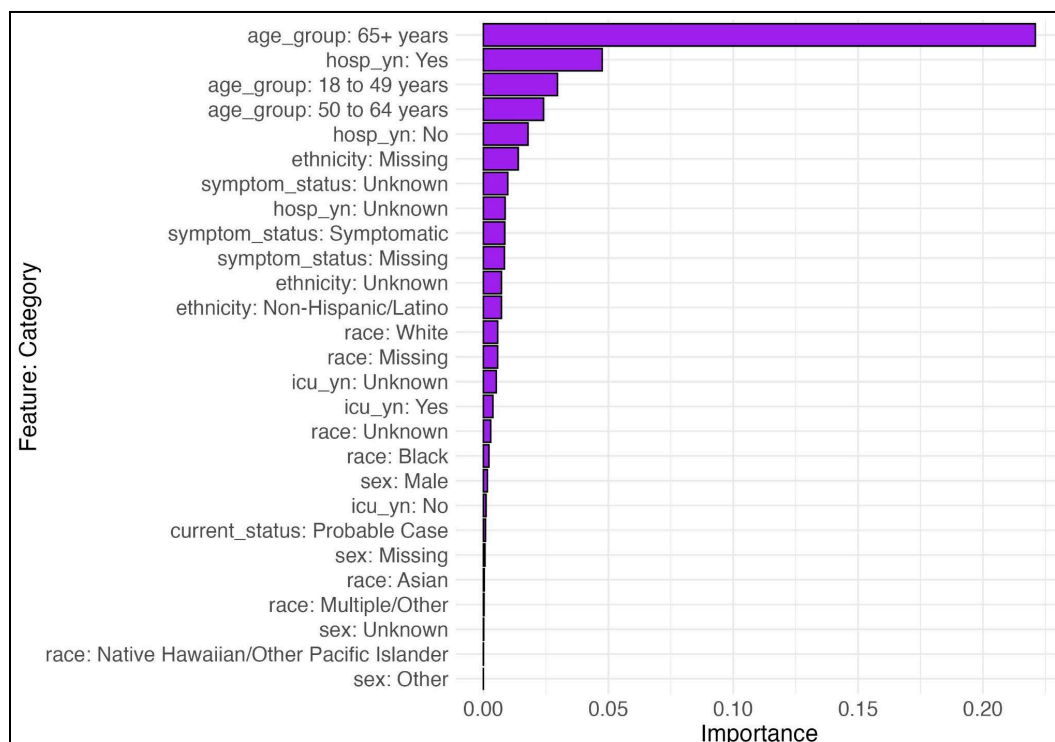
Predicted	Yes	41,367	4,505
	No	273,219	173
		No	Yes
		Actual	

While both of the random forest models achieved similar results, the sensitivity-optimized primary model is more appropriate for the stated objective. This model may incorrectly predict more patients; however, it will also draw more attention to those who are predicted to die as a result of COVID, which is favorable. Ultimately, the primary model allows for the correct identification of patients who will die with a 96.30 percent accuracy for only a slight increase in overall misclassification.

PFI scores are reported in Figure 4, with the importance being the percentage reduction in model accuracy from shuffling the given feature. Among the encoded features, the 65-plus age group had the largest importance. When breaking the connection between this encoded feature

and the death status, the primary model resulted in roughly a 22 percent reduction in accuracy, indicating the 65-plus age group contributed substantially to determining the death status of patients. Additionally, when a patient's hospitalization status was shuffled, prediction accuracy decreased by nearly 5 percent, making it a moderately predictive feature.

Figure 4: Permutation Feature Importance Scores by Feature-Category Combination



When comparing PFI results to the conditional probabilities of death found during exploratory data analysis, it became clear that the 65-plus age group contributed substantially to determining a patient's death status and had a large conditional probability of death. This indicates that age is strongly associated with COVID-19 mortality and is a key predictive factor in the model, making it a potential risk factor. A similar analysis tells us that being hospitalized is also associated with COVID-related death.

In contrast, whether a person was admitted into the ICU contributed relatively little predictive value to our model, as given by its PFI. Based on this factor's conditional probability

of death, it was expected to be a large contributor to model performance. In general, many of the observed relationships align with clinical intuition, but the presence of multicollinearity among features could be impacting certain features' relative importances, such as ICU admission.

V. Conclusion

This analysis created and evaluated two random forest models to predict and identify features associated with COVID-related death using patient records from CDC surveillance data. Due to the size of the data, the presence of messy features, and the highly imbalanced classes, many preprocessing steps were taken. Ultimately, the random forest models were trained on class-balanced data and evaluated on a stratified test dataset to ensure that model performance was subject to the true distribution of patient cases. Because the underlying distribution of patient cases has significantly more survivors than deceased patients, a naive baseline model was able to achieve an accuracy of 98.53 percent. While this demonstrated that simple models could easily predict the death status with a high degree of accuracy, it ultimately showed that if left up to accuracy-based modeling, the patients who die will be predicted incorrectly——this is the worst-case scenario in a healthcare context.

The random forest models substantially improved the predictive performance on patients who inevitably died, demonstrating the need for more advanced modelling beyond the naive model. While the accuracy-based and sensitivity-based models were optimized differently, they resulted in very similar predictions. The primary, sensitivity-based, random forest model allows for the accurate prediction of death for 86.99 percent of patients while correctly predicting 96.30 percent of those who actually died, making it highly beneficial in the healthcare field.

Upon analysis of permutation feature scores and conditional probabilities of death given specific features, certain factors were revealed to be associated with COVID-related death. Age, specifically individuals 65 and older, was the most influential factor in predicting death and had higher rates of death than those in other age groups. The hospitalization status of a patient played a similar but less influential role than age. Certain features, such as ICU admission, were expected to be more influential, but their importance was likely reduced due to the high correlation between features.

Ultimately, the results of the primary random forest model demonstrated that machine learning techniques can be used to predict COVID-related death using few patient features, with the most optimal clinical results being achieved when model sensitivity is the primary objective. Furthermore, combining model-based importance measures with exploratory conditional probability analysis provides a pathway to identifying potential risk factors. Because the models resulted in high predictive performance, this work would benefit most from extended analysis in risk-factor determination. Improvements could be made by examining how feature correlation impacts feature importance in this setting or by including causal analysis models that directly provide risk factors.

References

Chan B. (January 9, 2026). *Permutation Feature Importance Explained*. Medium.

<https://medium.com/data-science-explained/permutation-feature-importance-explained-76d1398b1c6f>

Rdocumentation (n.d.). *trainControl: Control parameters for train*.

<https://www.rdocumentation.org/packages/caret/versions/6.0-92/topics/trainControl>

U.S. Centers for Disease Control and Prevention. (February 23, 2025). *COVID-19 Case Surveillance Public Use Data with Geography*.

https://data.cdc.gov/Case-Surveillance/COVID-19-Case-Surveillance-Public-Use-Data-with-Ge/n8mc-b4w4/about_data

U.S. Centers for Disease Control and Prevention. (April 16, 2026). *Provisional COVID-19 Mortality Surveillance*. <https://www.cdc.gov/nchs/nvss/vsrr/covid19/index.htm>

Wright M. (n.d.). *ranger: Ranger*. Rdocumentation.

<https://www.rdocumentation.org/packages/ranger/versions/0.16.0/topics/ranger>